

CSA10 - Software Engineering

Github Lab Experiments

Experiment 30: Implement a Feature Using Fork and Pull Request Workflow

Aim

To fork a repository, create a new branch, implement a feature, and submit a Pull Request to the original repository.

Requirements

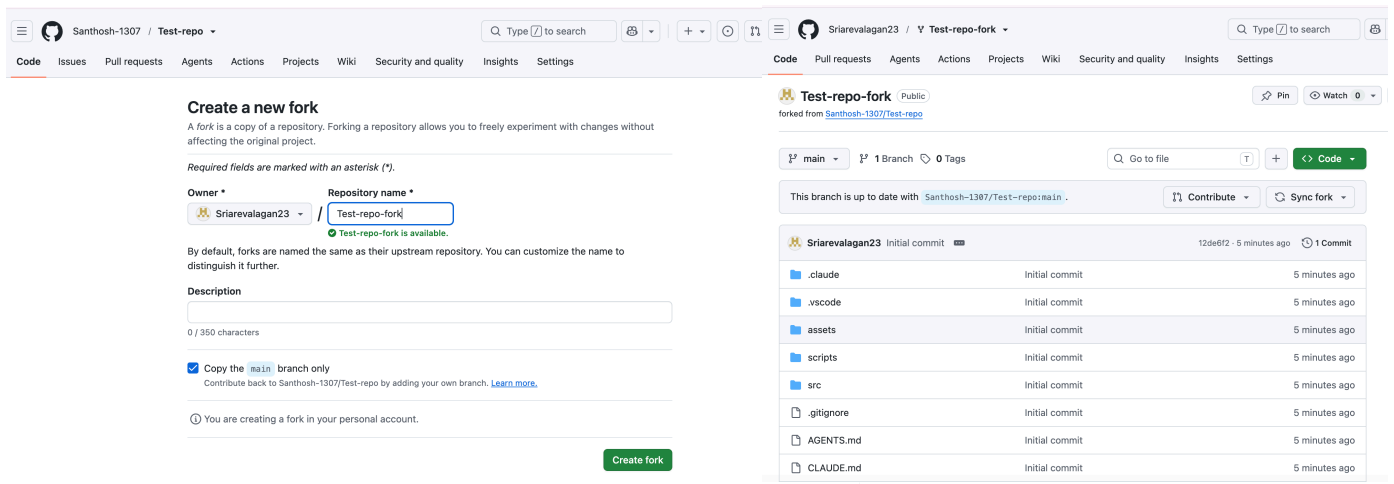
- Git installed
- GitHub account
- VS Code (or any text editor)

Procedure

Step 1: Fork the Repository

- Fork the required GitHub repository to your GitHub account.

Screenshot: Forked repository.



Step 2: Clone the Forked Repository

Bash

```
git clone <repository-url>
```

```
cd <repository-name>
```

Screenshot: Repository cloned.

```
Desktop — -zsh — 80x24

Last login: Tue Jul 21 22:18:52 on ttys000
[cd desktop]
srinidhi@Srinidhis-Laptop ~ % cd desktop
[srinidhi@Srinidhis-Laptop desktop % git clone https://github.com/Sriarevalagan23/Test-repo-fork
Cloning into 'Test-repo-fork'...
remote: Enumerating objects: 70, done.
remote: Counting objects: 100% (70/70), done.
remote: Compressing objects: 100% (63/63), done.
remote: Total 70 (delta 1), reused 70 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (70/70), 1.41 MiB | 6.25 MiB/s, done.
Resolving deltas: 100% (1/1), done.
srinidhi@Srinidhis-Laptop desktop %
```

Step 3: Create a New Branch

Bash

```
[srinidhi@Srinidhis-Laptop desktop % cd Test-repo-fork
[srinidhi@Srinidhis-Laptop Test-repo-fork % git checkout -b update-readme
Switched to a new branch 'update-readme'
srinidhi@Srinidhis-Laptop Test-repo-fork %
```

`git checkout -b feature-branch`

Screenshot: Branch created.

Step 4: Implement the Feature

- Make the required changes to the project files.
- Save the changes.
- Commit and push the branch.

Bash

`git add .`

`git commit -m "Added new feature"`

```
[srinidhi@Srinidhis-Laptop Test-repo-fork % git add .
[srinidhi@Srinidhis-Laptop Test-repo-fork % git commit -m "Updated README with pr
object description"
[update-readme 5e58284] Updated README with project description
1 file changed, 3 insertions(+)
[srinidhi@Srinidhis-Laptop Test-repo-fork % git push origin update-readme
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 385 bytes | 385.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'update-readme' on GitHub by visiting:
remote:   https://github.com/Sriarevalagan23/Test-repo-fork/pull/new/update-r
eadme
remote:
To https://github.com/Sriarevalagan23/Test-repo-fork
 * [new branch]   update-readme -> update-readme
srinidhi@Srinidhis-Laptop Test-repo-fork %
```

`git push origin feature-branch`

Screenshot: Changes pushed.

Step 5: Create a Pull Request

- Create a Pull Request from your forked repository to the original repository.

Screenshot: Pull Request created.

Open a pull request
Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).

base repository: Santhosh-1307/Test-repo base: main ← head repository: Sriarevalagan23/Test-repo-f... compare: update-readme

✓ **Able to merge.** These branches can be automatically merged.

Add a title *
Updated README with project description

Add a description

Write Preview H B I

new changes have been made in the read me file.

☐ Markdown is supported ☐ Paste, drop, or click to add files

☒ Allow edits by maintainers **Create pull request**

Reviewers
No reviews

Assignees
No one—[assign yourself](#)

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

Result

Successfully forked a repository, created a feature branch, implemented the required changes, pushed them to GitHub, and submitted a Pull Request to the original repository.